Chapter 11

Message Integrity
and
Message Authentication

11-1 MESSAGE INTEGRITY

The cryptography systems that we have studied so far provide secrecy, or confidentiality, but not integrity. However, there are occasions where we may not even need secrecy but instead must have integrity.

Topics discussed in this section:

11.1 Document and Fingerprint
11.2 Message and Message Digest
11.3 Difference
11.4 Checking Integrity
11.5 Cryptographic Hash Function Criteria

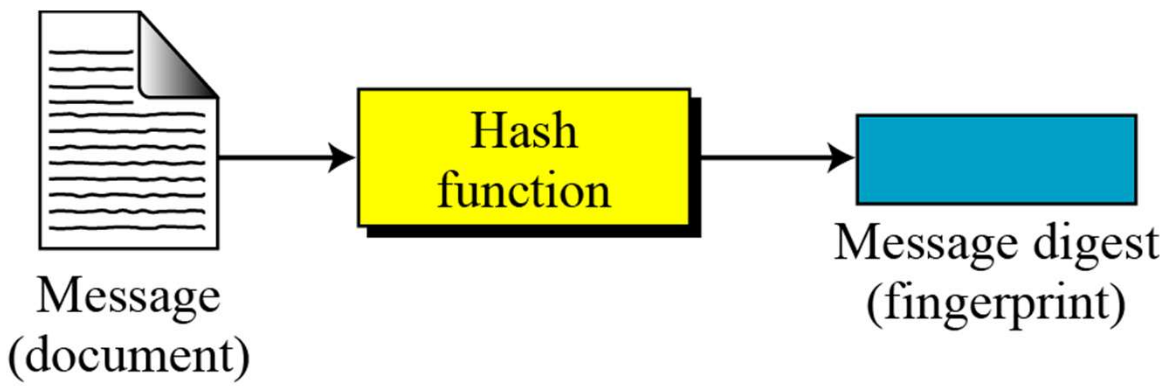
11.1.1 Document and Fingerprint

One way to preserve the integrity of a document is through the use of a fingerprint. If Alice needs to be sure that the contents of her document will not be changed, she can put her fingerprint at the bottom of the document.

11.1.2 Message and Message Digest

The electronic equivalent of the document and fingerprint pair is the message and digest pair.

Figure 11.1 ***Message and digest***



11.1.3 Difference

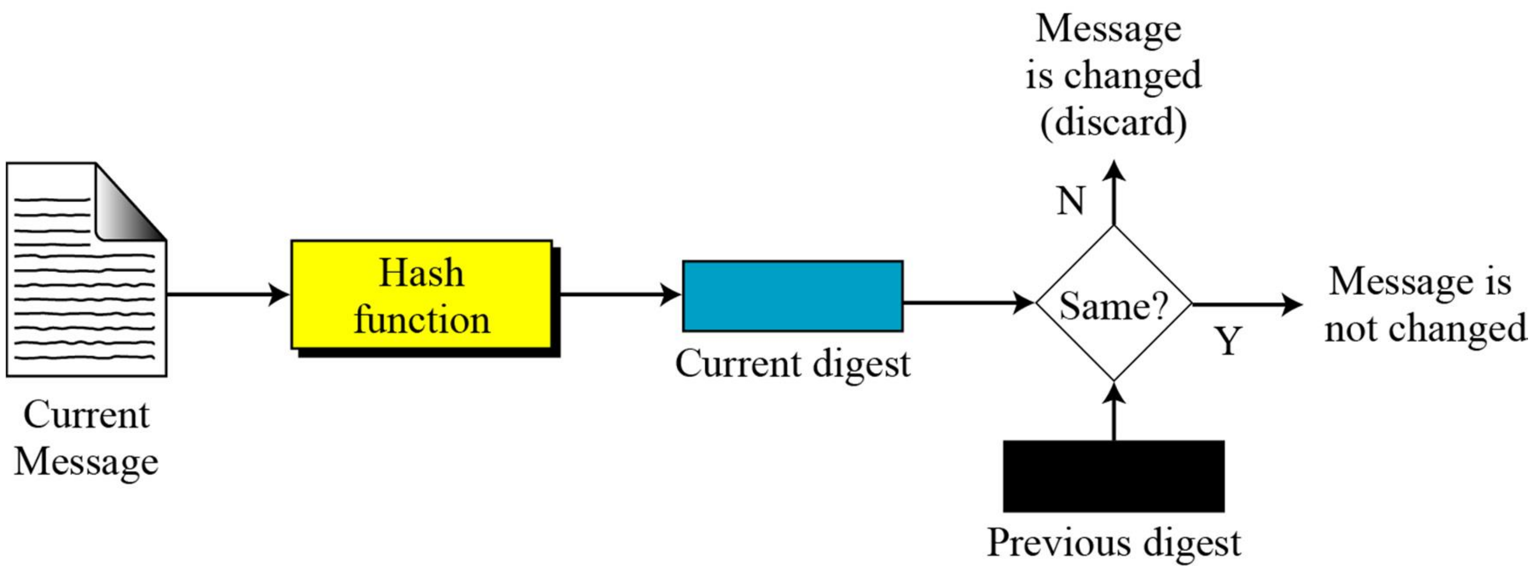
The two pairs (document / fingerprint) and (message / message digest) are similar, with some differences. The document and fingerprint are physically linked together. The message and message digest can be unlinked separately, and, most importantly, the message digest needs to be safe from change.

Note

The message digest needs to be safe from change.

11.1.4 Checking Integrity

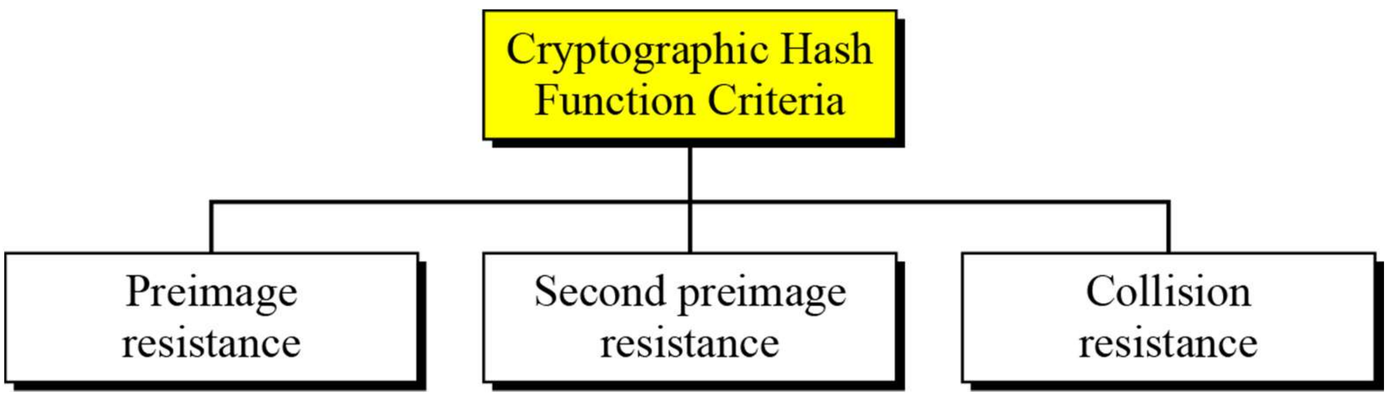
Figure 11.2 ***Checking integrity***



11.1.5 Cryptographic Hash Function Criteria

A cryptographic hash function must satisfy three criteria: preimage resistance, second preimage resistance, and collision resistance.

Figure 11.3 *Criteria of a cryptographic hash function*



Cryptographic hash functions must meet specific criteria to ensure their effectiveness in various security applications. Here are the key properties or criteria of a cryptographic hash function:

1. **Collision Resistance**:

- - Definition: It should be computationally infeasible to find two different inputs that produce the same hash value.
- - Importance: Collision resistance is crucial to prevent attackers from creating different inputs with identical hash values, preserving the integrity of the hashing process.

2. **Preimage Resistance (One-Way Function):**

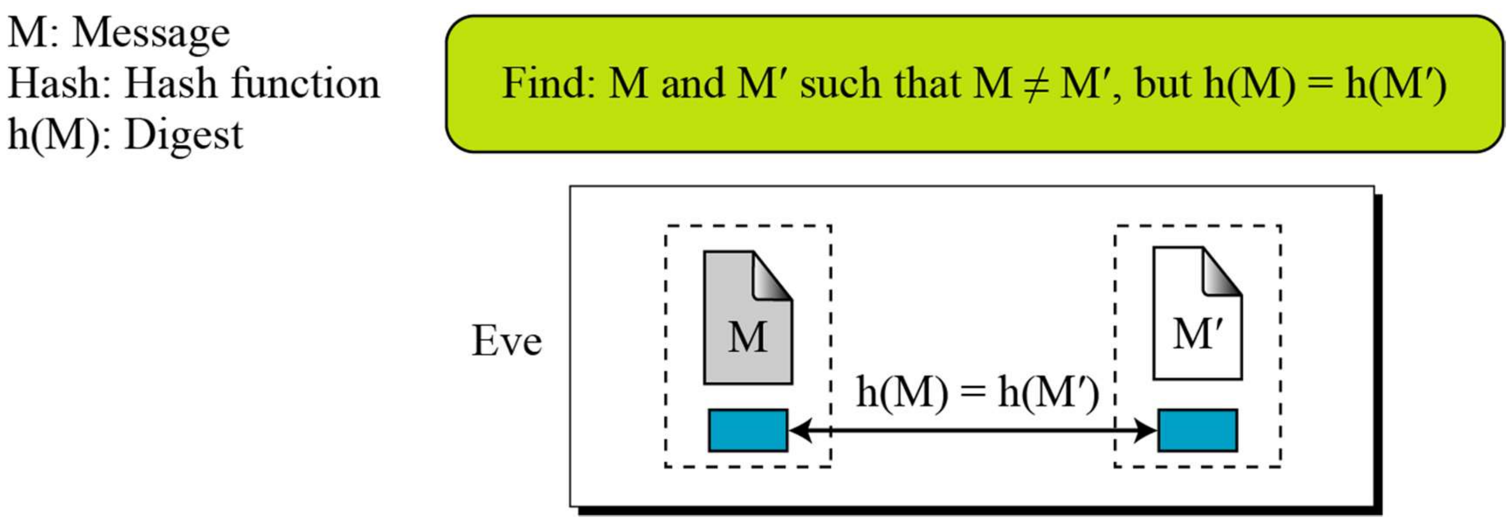
- - Definition: Given a hash value, it should be computationally infeasible to find any input that produces that hash value.
- - Importance: This property ensures that it's difficult for an attacker to reverse the hash function and obtain the original input, enhancing security.

Collision Resistance

Collision Attack

Given: none **Find: $M' \neq M$ such that $h(M) = h(M')$**

Figure 11.6 ***Collision***



Preimage Resistance

Preimage Attack

Given: y = h(M) **Find: M′ such that y = h(M′)**

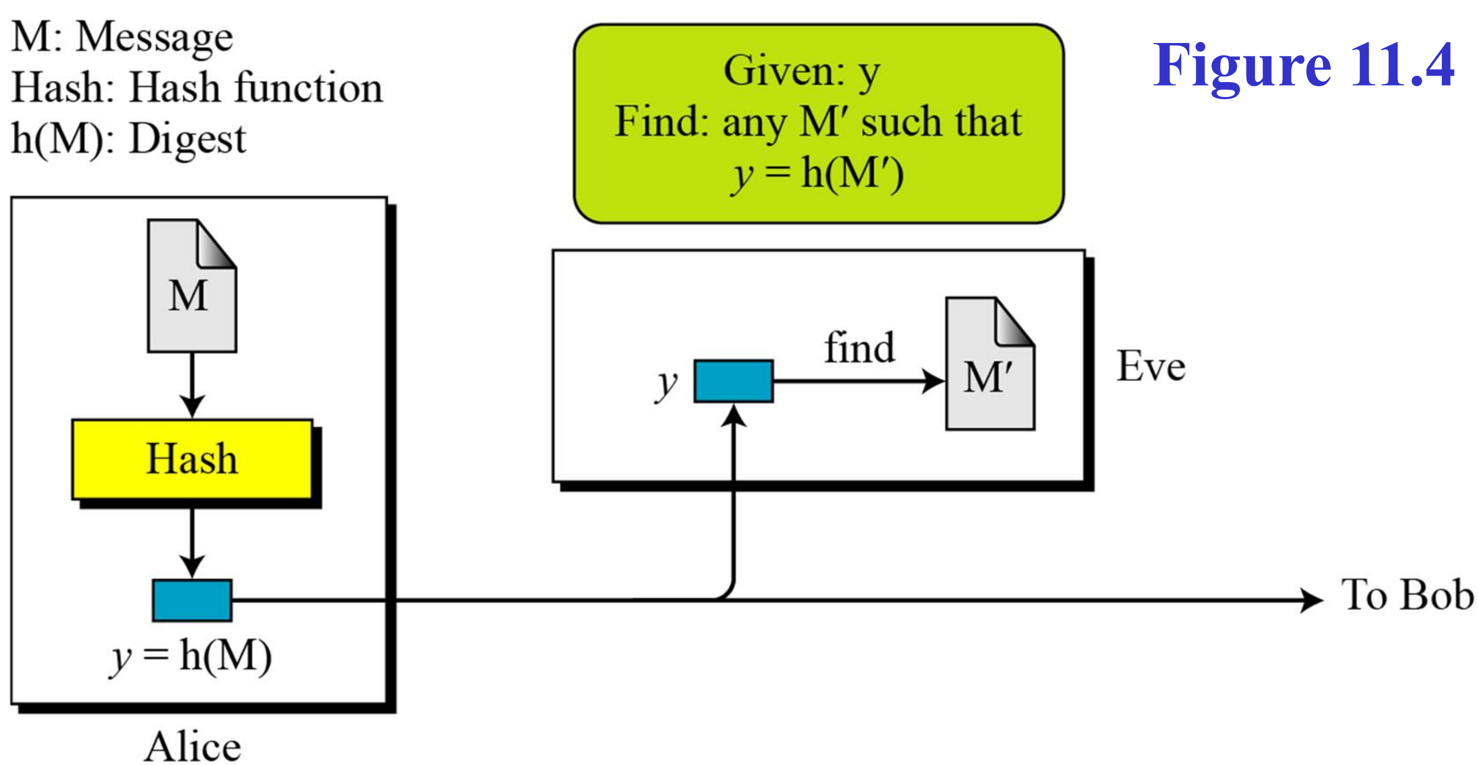


Figure 11.4 ***Preimage***

3. **Second Preimage Resistance**:

- \- Definition: Given an input and its hash value, it should be computationally infeasible to find a different input that produces the same hash value.
- \- Importance: Second preimage resistance prevents attackers from finding alternative inputs that map to the same hash value as a known input.

4. **Avalanche Effect**:

- \- Definition: A small change in the input should result in a significantly different hash value.
- \- Importance: The avalanche effect ensures that similar inputs produce vastly different hash values, making it difficult for attackers to identify patterns or relationships between inputs and outputs..

Second Preimage Resistance

Second Preimage Attack

Given: M and h(M) **Find: M′ ≠ M such that h(M) = h(M′)**

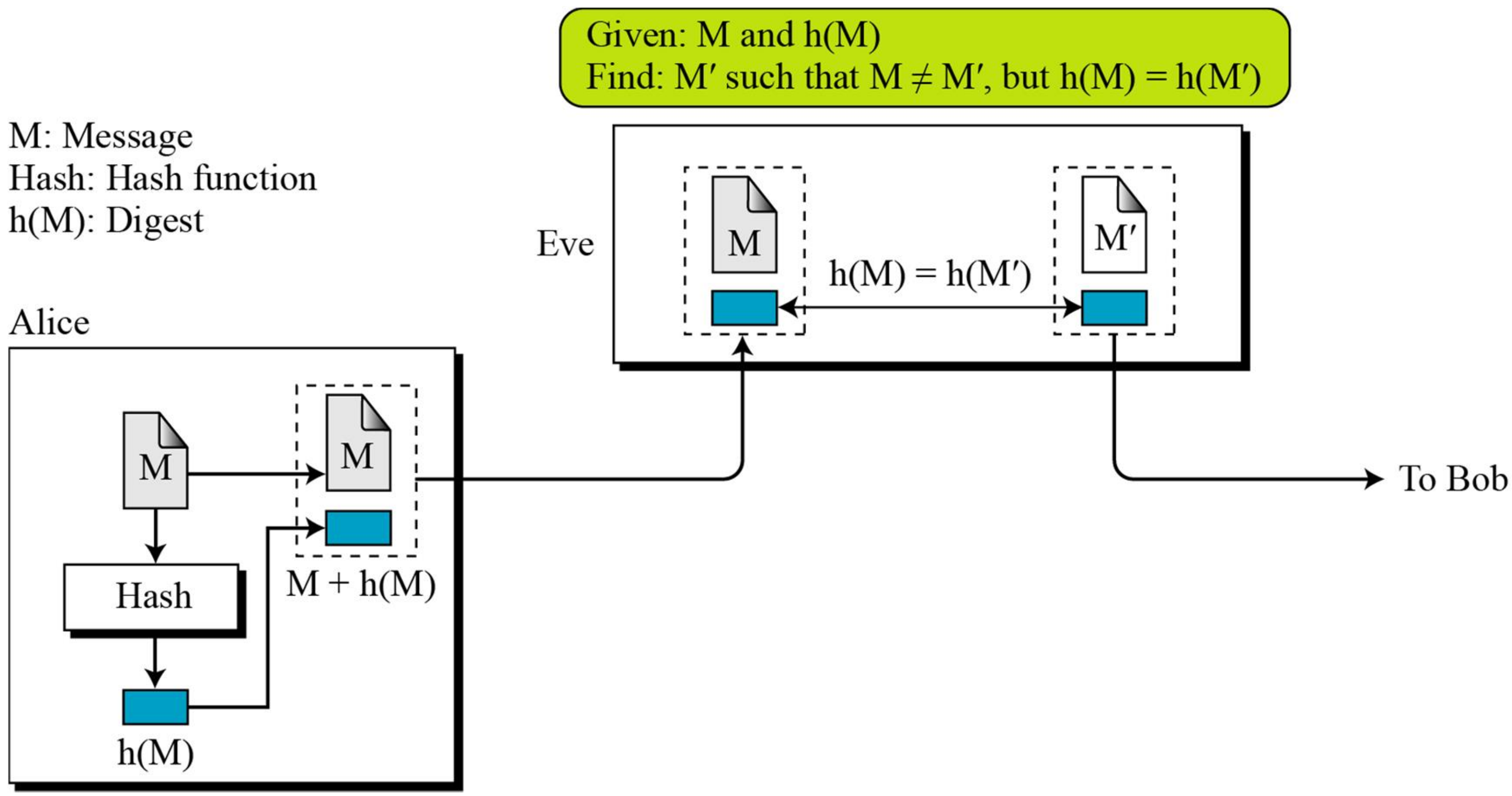


Figure 11.5 ***Second preimage***

5. **Deterministic**:

- \- Definition: For a given input, the hash function must produce the same hash value every time it is applied.
- \- Importance: Deterministic behavior is essential for consistency in cryptographic operations and ensures the predictability of hash values.

6. **Efficiency**:

- \- Definition: The hash function should be computationally efficient to compute the hash value for any given input.
- \- Importance: Efficiency is crucial for practical implementation in various cryptographic protocols and systems.

7. **Resistance to Birthday Attacks**:

- \- Definition: The hash function should resist attacks that exploit the birthday paradox, where the probability of two different inputs having the same hash value becomes non-negligible.
- \- Importance: Birthday attacks aim to find collisions more efficiently than exhaustive search and pose a threat to hash functions if not properly addressed.

Ensuring these properties helps to create robust cryptographic hash functions that are resistant to various attacks and provide a foundation for secure communication and data integrity

Can we use a conventional lossless compression method such as *StuffIt* as a cryptographic hash function?

Solution

We cannot. A lossless compression method creates a compressed message that is reversible.

Can we use a checksum function as a cryptographic hash function?

Solution

We cannot. A checksum function is not preimage resistant, Eve may find several messages whose checksum matches the given one.